

MPPI Covariance Adaptation Proof

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MPPI with Covariance Adaptation

Scenario

Suppose an MPPI-controlled robot is in a trap, which we denote as the set $\mathcal{T}$. In a trapped set, the probability of the robot sampling a control sequence which allows it to escape, p_{esc} , is approximately zero. Therefore, it finds a local minimizer of the cost function $\mathcal{J}(\mathbf{x}, \mathbf{u})$ which is greater than the globally minimizing trajectory. We aim to increase p_{esc} while maintaining computational efficiency.

Claim and Proof

Given a robot with dynamics $\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k)$, $\mathbf{x}_k \in \mathcal{T} \subset \mathbb{R}^{n_x}$, $\mathbf{u}_k \in \mathbb{R}^{n_u}$ and an MPPI control algorithm which finds the optimal control input $\mathbf{u}_k^*$ according to

$$\mathbf{u}_k^* = \sum_{m=1}^M w_m \delta \mathbf{u}_{k,m}$$

$$\text{where } w_m = \frac{\exp\left(-\frac{1}{\lambda}(\mathcal{J}_m - \tilde{S})\right)}{\sum_{m=1}^M \exp\left(-\frac{1}{\lambda}(\mathcal{J}_m - \tilde{S})\right)}, \quad \delta \mathbf{u}_{k,m} \sim \mathcal{N}(0, \Sigma_0), \quad \Sigma_0 \succ 0$$

$$\text{and } \mathcal{J}_m = \mathcal{J}(\underline{\mathbf{x}}^{(m)}, \underline{\mathbf{u}}^{(m)}) = \phi(\mathbf{x}_T) + \sum_{i=k}^{T+k-1} \mathcal{L}(\mathbf{x}_i, \mathbf{u}_i + \delta \mathbf{u}_i^{(m)}) \text{ with } \tilde{S} = \min\{\mathcal{J}_1, \dots, \mathcal{J}_M\}$$

Additionally, it is known that for the entrapped space the robot is in,

$$\min_{\mathbf{x}_k, \mathcal{T}} \mathcal{J} > \min_{\mathbf{x}_k, \mathcal{E}} \mathcal{J}$$

We claim that for any $\Sigma > \Sigma_0$, the probability of escaping the trapped set, p_{esc} increases.

Remark 1: For any trajectory $\underline{\mathbf{x}}$ with some initial condition $\mathbf{x}_0$, there exists a unique control sequence $\underline{\mathbf{u}}$ such that $\underline{\mathbf{x}}$ is generated by propagating through the system dynamics.

Proof. We first let $\underline{\mathbf{x}}_e^*$ denote the escape trajectory. By remark 1, there must exist a control sequence $\underline{\mathbf{u}}_e^*$ which yields this escape trajectory. Since MPPI perturbs around some nominal control sequence $\underline{\mathbf{u}}_{nom}$, we derive that there exists a control perturbation $\delta\mathbf{u}^*$ such that

$$\begin{aligned}\underline{\mathbf{u}}_{nom} + \delta\mathbf{u}^* &\approx \underline{\mathbf{u}}_e^* \\ \iff \delta\mathbf{u}^* &\approx \underline{\mathbf{u}}_e^* - \underline{\mathbf{u}}_{nom} \triangleq \Delta\underline{\mathbf{u}}^* \\ \iff \delta\mathbf{u}^* - \Delta\underline{\mathbf{u}}^* &\approx 0\end{aligned}$$

The escape probability, p_{esc} , is mathematically defined as

$$p_{esc} = \prod_{i=k}^{T+k-1} P(\|\delta\mathbf{u}_{k,m} - \Delta\mathbf{u}_k\| \leq \varepsilon)$$

where ε is a very small computational error tolerance. Recall that the perturbations are sampled from a Mixture-of-Gaussians (MoG) distribution with zero mean and a positive definite covariance matrix Σ with equal eigenvalues. We therefore write $\Sigma = \sigma^2 I_{n_u}$, where I_{n_u} is the identity matrix of dimension n_u . The probability density function is defined as follows.

$$p(\delta\mathbf{u}_{k,m}; \sigma) = \frac{\alpha}{\sigma^{n_u}} \exp\left(-\frac{\|\delta\mathbf{u}_{k,m}\|^2}{2\sigma^2}\right)$$

where α is a proportionality constant designed to satisfy the law of total probability. To determine where increasing σ increases the probability density function, we take the derivative of the Gaussian PDF with respect to σ , which yields

$$\begin{aligned}\frac{\partial p}{\partial \sigma} &= \alpha \left[-\frac{n_u}{\sigma \cdot \sigma^{n_u}} \exp\left(-\frac{\|\delta\mathbf{u}_{k,m}\|^2}{2\sigma^2}\right) + \frac{\|\delta\mathbf{u}_{k,m}\|^2}{\sigma^3 \cdot \sigma^{n_u}} \exp\left(-\frac{\|\delta\mathbf{u}_{k,m}\|^2}{2\sigma^2}\right) \right] \\ &= \frac{\alpha}{\sigma^{n_u+1}} \left[\frac{\|\delta\mathbf{u}_{k,m}\|^2}{\sigma^2} - n_u \right] \exp\left(-\frac{\|\delta\mathbf{u}_{k,m}\|^2}{2\sigma^2}\right) > 0\end{aligned}$$

Since exponential functions and σ are both strictly positive for all $\delta\mathbf{u}_{k,m}$, we are not interested in their effect. Therefore, we ascertain that the above inequality

is equivalent to

$$\begin{aligned}
& \frac{\|\delta \mathbf{u}_{k,m}\|^2}{\sigma^2} - n_u > 0 \\
& \iff \|\delta \mathbf{u}_{k,m}\|^2 > \sigma^2 n_u \\
& \iff \|\delta \mathbf{u}_{k,m}\| > \sigma \sqrt{n_u}
\end{aligned}$$

Therefore, for input perturbations with large Euclidean norms, increasing the perturbation covariance increases the probability density at those perturbations, which ultimately increases the probability of sampling within a neighborhood of this perturbation. This is desired for entrapment scenarios where the desired control sequence deviates heavily from the nominal control sequence, yielding $\|\Delta \mathbf{u}^*\| \gg \varepsilon$. Therefore, the probability $P(\|\delta \mathbf{u}_{k,m} - \Delta \mathbf{u}_k\| \leq \varepsilon)$ increases, which must increase p_{esc} for any $\Sigma > \Sigma_0$.

From a controls perspective, we can additionally claim that MPPI has a higher probability of deriving an optimal control input which allows the robot to escape. The entrapment has a local minima of the performance index which is significantly greater than the global minima of the performance index for the current state of the robot. Suppose a control sequence $\tilde{\mathbf{u}}^* \approx \mathbf{u}^*$ is sampled by MPPI, which yields an approximately pseudo-optimal trajectory $\tilde{\mathbf{x}}^*$. We prove this claim below.

$$\begin{aligned}
& \min_{\mathbf{x}_k, \mathcal{T}} \mathcal{J} > \min_{\mathbf{x}_k, \mathcal{E}} \mathcal{J} \\
& \implies \mathcal{J}(\tilde{\mathbf{x}}^*, \tilde{\mathbf{u}}^*) < \mathcal{J}(\mathbf{x}, \mathbf{u}) \quad \forall \mathbf{x} \in \mathcal{X}_{\mathcal{T}}, \mathbf{u} \in \mathcal{U}_{\mathcal{T}} \\
& \quad \therefore \tilde{S} \approx \mathcal{J}(\tilde{\mathbf{x}}^*, \tilde{\mathbf{u}}^*) \iff w^* \approx \max\{w_1, \dots, w_m\}
\end{aligned}$$

Based on this result, we ascertain that the escape perturbation contributes the most to the optimal control input generated by MPPI. Therefore, for larger input covariance Σ , there will be a larger probability that MPPI finds a control input which allows the robot to escape the entrapment set. $\square$